



SOP #14

Sample Prep (WB)

1. Decide **how** to load. Possible options include on a total protein basis, by sample wet weight, by Chlorophyll A content, etc.
2. Calculate the ideal “single load” of protein extract based on requirement (ChlA/wet weight/total protein) and the assay outcome for that requirement. In our example in, protein loads are prepared on a total protein basis, and calculations are done using the protein concentrations from the Biorad DC assay or Thermo Scientific BCA assay.
3. Our default ideal load volume is 10 uL because it fits into most well sizes. If the wells hold more, you may increase your load volume size. This is especially true if the protein assay results were very dilute - you may need to increase load volumes just to get enough buffer in your gel sample.
4. In an ideal 10 uL load, the 4 components would be:

- protein extract solution representing 1 ug (or 5 ug, or 30 ug) total protein _____ uL
- 10 % by volume 0.5 M DTT solution (final con. 0.05 M) = 1.0 uL
- 1X blue sample buffer (contains dye & glycerol & soap) to bring the solution to 10 uL total volume

Note: this means the 1X sample buffer will be less than 1X concentration in the final volume. That's OK. A 100 uL solution of 1x sample buffer is: 75 uL water + 25 uL 4XSB LDS sample buffer (purple solution)

For example: 1 uL protein extract diluted with 8 uL of 1XSB + 1 uL DTT. While the final concentration of buffer components will be slightly less than 1X, it should still denature & sink into the well efficiently. The sample buffer also contains more LDS soap to coat the proteins (giving them a negative charge). At least 20% of your sample volume should be 1X purple sample buffer.

5. Once the ideal single load is calculated, multiply by a factor of 1.5X (or 2X) for a single load so that you make excess.
6. Creating a standard curve of known protein concentrations is required for any sort of quantitative work. Since the protein standards (for PsbA, RbcL, NifH for example) are already diluted in 1X sample buffer + 10% DTT, dilutions are always done in more 1X sample buffer +10% DTT. For 100 uL: 65 uL water + 25 uL 4X sample buffer + 10 uL DTT (0.5M)

Standard curves can either be created by pipetting protein standards independently as dilutions, or as a serial dilution.

7. Heat all tubes 70 C for 5 minutes. Spin to collect condensation. Pipette up-down to mix right before loading. Tubes can also be stored in -20 C freezer until needed. When ready to load, thaw samples on ice, heat at 70 C for 2 minutes only. Spin, pipette to mix and load.